

This exam has 6 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Semantic entailment (5 + 5 points)

- (a) Suppose the semantic entailment $\phi_1 \models \phi_2$ holds. Can we conclude that then the semantic entailment $(\phi_1 \rightarrow \psi) \models (\phi_2 \rightarrow \psi)$ holds? If so, show this by means of a logical argumentation or a truth table; if not, give a counterexample.

Solution: The semantic entailment does not hold in general. Take ϕ_1 to be F, ϕ_2 to be T, and ψ to be F. Then $\phi_1 \models \phi_2$. But, since $\phi_1 \rightarrow \psi$ is T and $\phi_2 \rightarrow \psi$ is F, $(\phi_1 \rightarrow \psi) \models (\phi_2 \rightarrow \psi)$ does not.

- (b) Suppose the semantic entailment $\psi_1 \models \psi_2$ holds. Can we conclude that then the semantic entailment $(\phi \rightarrow \psi_1) \models (\phi \rightarrow \psi_2)$ holds? If so, show this by means of a logical argumentation or a truth table; if not, give a counterexample.

Solution: We can conclude that the semantic entailment holds. The logical argumentation is that $\psi_1 \models \psi_2$ implies that $\psi_1 \rightarrow \psi_2$ is a tautology. So if $\phi \rightarrow \psi_1$ is T, then (by transitivity of implication) $\phi \rightarrow \psi_2$ is T.

The argumentation via a truth table is:

ϕ	ψ_1	ψ_2	$\phi \rightarrow \psi_1$	$\phi \rightarrow \psi_2$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	F
F	T	T	T	T
F	T	F	T	T
F	F	T	T	T
F	F	F	T	T

Since $\psi_1 \models \psi_2$, rows 2 and 6 where ψ_1 is T and ψ_2 is F are ruled out. In the remaining rows, $\phi \rightarrow \psi_1$ is F or $\phi \rightarrow \psi_2$ is T.

2. Island puzzle (8 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). Suppose an islander A would say: “If I am a liar, then I am a truth speaker.”

Determine via a truth table whether A would be a truth speaker, a liar, both options are possible, or this statement is impossible on the island.

Solution: The formalization of the statement is $t_A \leftrightarrow (\neg t_A \rightarrow t_A)$.

t_A	$\neg t_A$	$\neg t_A \rightarrow t_A$	$t_A \leftrightarrow (\neg t_A \rightarrow t_A)$
T	F	T	T
F	T	F	T

Since the fourth column contains two T values, both options are possible: A can be a truth speaker or a liar.

3. Conjunctive normal form (8 points)

Give the truth table of the DNF $(p \wedge q) \vee r$, and use it to construct a formula in CNF that is semantically equivalent.

Solution:

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$
T	T	T	T	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	T
F	F	F	F	F

Since only the fourth, sixth and eighth row of the fifth column contain F, the resulting CNF is

$$(\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee r)$$

4. CNF algorithm (10 points)

Apply the algorithm CNF to turn the formula $\neg(p \rightarrow q) \vee (\neg p \wedge q)$ into CNF.

Solution:

$$\begin{aligned}
\neg(p \rightarrow q) \vee (\neg p \wedge q) &\rightsquigarrow_{\text{IMPL-FREE}} \neg(\neg p \vee q) \vee (\neg p \wedge q) \\
&\rightsquigarrow_{\text{NNF}} (\neg\neg p \wedge \neg q) \vee (\neg p \wedge q) \\
&\rightsquigarrow_{\text{NNF}} (p \wedge \neg q) \vee (\neg p \wedge q) \\
&\rightsquigarrow_{\text{DISTR}} ((p \wedge \neg q) \vee \neg p) \wedge ((p \wedge \neg q) \vee q) \\
&\rightsquigarrow_{\text{DISTR}} (p \vee \neg p) \wedge (\neg q \vee \neg p) \wedge ((p \wedge \neg q) \vee q) \\
&\rightsquigarrow_{\text{DISTR}} (p \vee \neg p) \wedge (\neg q \vee \neg p) \wedge (p \vee q) \wedge (\neg q \vee q)
\end{aligned}$$

5. DPLL procedure (9 points)

Apply the DPLL procedure to the CNF $(\neg p \vee \neg q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$, to check whether it is satisfiable.

Solution: First try p is $\top$:

$$(\neg \top \vee \neg q) \wedge (\neg \top \vee q) \wedge (\top \vee \neg q)$$

This reduces to $q \wedge \neg q$. Both attempting $\top$ or $\perp$ for q yields $\perp$, so the temporary verdict is “unsatisfiable”.

Now try p is $\perp$:

$$(\neg \perp \vee \neg q) \wedge (\neg \perp \vee q) \wedge (\perp \vee \neg q)$$

This reduces to $\neg q$. Attempting $\top$ for q yields $\perp$, giving the temporary verdict “unsatisfiable”. Finally, attempting $\perp$ for q yields $\top$. So the final verdict is “satisfiable”, if both p and q have the value $\perp$.

6. Sets (6 + 9 points)

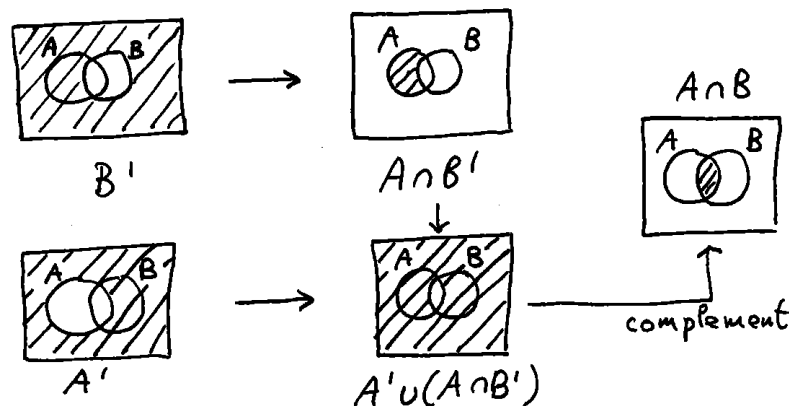
In this task, we consider the following set-theoretic equality:

$$(A' \cup (A \cap B'))' = A \cap B$$

- Draw Venn diagrams for the two sets $(A' \cup (A \cap B'))'$ and $A \cap B$, depicting clearly which area corresponds to the set, and how the set is constructed from A and B using the fundamental set operations. Use your Venn-diagrams to conclude that the set-theoretic equality displayed above holds.
- Prove the set-theoretic equality displayed above (for all sets A and B) using the rules of the algebra of sets.

Solution:

- The following Venn-diagrams confirm that the equality holds:



$$\begin{aligned}
(b) \quad (A' \cup (A \cap B'))' &= (A')' \cap (A \cap B')' && \text{(De Morgan)} \\
&= A \cap (A \cap B')' && \text{(involution)} \\
&= A \cap (A' \cup (B')') && \text{(De Morgan)} \\
&= A \cap (A' \cup B) && \text{(involution)} \\
&= (A \cap A') \cup (A \cap B) && \text{(distributivity)} \\
&= \emptyset \cup (A \cap B) && \text{(complement)} \\
&= A \cap B && \text{(identity)}.
\end{aligned}$$

7. Relations (4 + 6 + 6 points)

Consider the relations R and S , both in the set $V := \{1, 2, 3, 4\}$, given by the following matrix representations:

$$R: \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \qquad S: \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

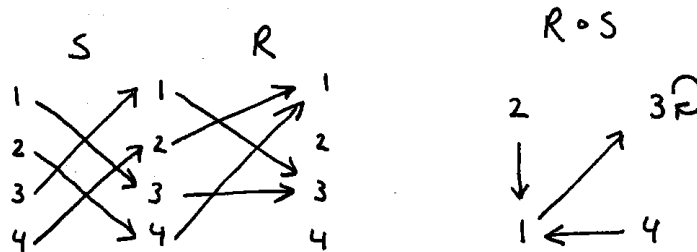
(rows and columns are numbered in the order 1, 2, 3, 4)

- Is the relation S reflexive? transitive? symmetric? anti-symmetric? Briefly explain your answers.
- Explicitly list all the elements of $R \circ S$ in the curly-bracket notation and draw the directed graph representation of $R \circ S$.
- Find a relation in the set V that is *neither* symmetric *nor* anti-symmetric, and draw its directed graph representation.

Solution:

- Reflexive: no, no element relates to itself.
Transitive: no, we have $1 S 3$ and $3 S 1$, but not $1 S 1$.
Symmetric: yes, whenever an element x relates to y , y also relates to x .
Anti-symmetric: no, we have $1 S 3$ but also $3 S 1$.

- We draw a Venn representation of the composition $R \circ S$:

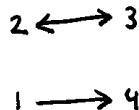


From the Venn representation we can read off all the pairs that together make up the composition:

$$R \circ S = \{\langle 1, 3 \rangle, \langle 2, 1 \rangle, \langle 3, 3 \rangle, \langle 4, 1 \rangle\}$$

We can now draw the directed graph representation; again see the picture above.

(c) An example is the relation depicted in the following directed graph:



8. Ordering relations (8 + 3 + 3 points)

In this task, we consider the Cartesian ordering relation in the set

$$A := \{a, b\}^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\},$$

induced by the alphabetic order on $\{a, b\}$. Note that we write **aba** instead of $\langle a, b, a \rangle$, and so on, to simplify the notation.

- (a) Use the algorithm that you have learned to construct the Hasse diagram that represents the Cartesian ordering relation on A . Explicitly write down the sets G_x and H_x obtained in the construction, then draw the Hasse diagram.

Let B be the subset of A that is given by

$$B := \{aaa, aab, aba, abb, bab\}.$$

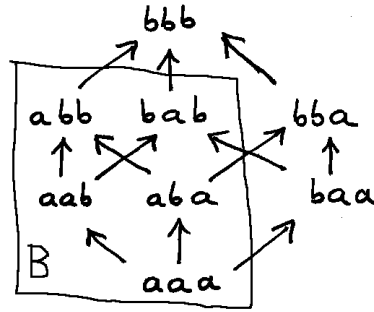
- (b) Does the set B have a smallest element according to the Cartesian ordering relation from part (a)? If so, please give this smallest element. If not, please list all the minimal elements of the set B .
- (c) Does the set B have a largest element according to the Cartesian ordering relation from part (a)? If so, please give this largest element. If not, please list all the maximal elements of the set B .

Solution:

(a) The algorithm gives the following sets G_x and H_x :

$G_{aaa} = \{aab, aba, abb, baa, bab, bba, bbb\}$	$H_{aaa} = \{aab, aba, baa\}$
$G_{aab} = \{abb, bab, bbb\}$	$H_{aab} = \{abb, bab\}$
$G_{aba} = \{abb, bba, bbb\}$	$H_{aba} = \{abb, bba\}$
$G_{baa} = \{bab, bba, bbb\}$	$H_{baa} = \{bab, bba\}$
$G_{abb} = \{bbb\}$	$H_{abb} = \{bbb\}$
$G_{bab} = \{bbb\}$	$H_{bab} = \{bbb\}$
$G_{bba} = \{bbb\}$	$H_{bba} = \{bbb\}$
$G_{bbb} = \emptyset$	$H_{bbb} = \emptyset$

The Hasse diagram therefore looks like this:



- (b) As the Hasse diagram shows, the smallest element in the set B is aaa .
- (c) As the Hasse diagram shows, the set B does not have a largest element. It has two maximal elements: abb and bab .